M

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE, ROCKET ASSISTED, XM548 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of rocket assisted artillery cartridge designated as Cartridge, 105MM, HE, Rocket Assisted, XM548 (including loading of the warhead and the motor body).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

PURCHASE DESCRIPTIONS

PA-PD-2791 - Primer, Percussion, XM108, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-453 - Inspection, Radiographic

FSC: 1315

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9209385 - Container, Ammunition, Plastic, M-, for Cartridge, 105MM, HE, RAP, XM548
9209387 - Crate, Wood, Open, Wirebound for 105MM (Howitzer) Ammunition, in Plastic Container XM578
9212376 - Cartridge, Semi-Fixed, 105MM, HE, Rocket Assisted, XM548 for Cannon M2A2, M103 or M137
9213636 - Container, Ammunition, Fiber, M-, for Cartridge, 105MM, HE, RAP, XM548
9213637 - Box, Packing, Ammunition, for Cartridge, 105MM, HE, RAP, XM548

PUBLICATIONS

ARMY

ET-9212376 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap.-Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.6).

3.2 Loaded Cartridge.-The loaded cartridge shall comply with all requirements specified on Drawing (dwg.) 9212376 and with requirements specified in applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.3 Cavitation in high explosive charge

3.3.1 Cracks and projected cavities.-Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.3.1. Cavities having a maximum projected length of 1/32 inch or less shall be disregarded.

TABLE I

	Segments (see Figure 1)			
	A*	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings (square inch (sq. in.))	1/64**	1/4	1/2	1/2
Projected length of any cavity excluding pipes cracks and annular rings (in.)	1/8	1/2	1/2	3/4
Piping cavities, max., projected area (sq. in.)	0	1/4	1/2	0
Piping cavities, max., projected width (in.)	0	1/4	1/4	0
Cracks, max., projected width (in.)	1/32	1/32	1/32	—
Annular rings, max., projected width (in.)	0	0	1/4	0

*Pertains to both composition B and filler E

** If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.3.2 Porous areas.-Porous areas shall be treated as cavities and shall be subjected to the restriction placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.3 Cracked charges.-Not more than four transverse cracks shall be permitted in any charge and not more than one transverse crack shall be permitted in Segment A.

3.3.4 Surface cavitation.-Cavities in Segment D of Figure 1 formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The max. dimension of any visible cavity in the base of the fuze well shall not exceed 1/4 inch in any direction and the total area of all such cavities shall not exceed 80 percent of the area of the fuze well base. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation). Disregard any cavitation in fuze well above Segment D.

3.3.5 Base separation.-There shall be no separation greater than 1/32 of an inch between the shell base and the filler E and likewise between the filler E and the explosive charge.

3.4 Loading of high explosive

3.4.1 Mating surfaces.-The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the warhead.

3.4.2 Solid explosives.-Solid explosives shall not be present in the warhead before loading or be added to the shell during pouring or cooling.

3.4.3 Coating.-Warhead having the interior wall imperfectly coated shall not be loaded. The interior surface of the warhead shall be dry when the high explosive is poured into the warhead.

3.5 Warhead disassembly torque.-The Warhead shall withstand a minimum disassembly torque of 200 foot pounds when tested as specified in 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.6 Primer.-The primer shall have the required number and location of flash holes as specified on the appropriate primer drawing.

3.7 Waterproofness (applicable to plastic containers).- The plastic container shall be waterproof as evidenced by absence of air leakage when tested as specified in 4.3.2.

3.8 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.3.3.

3.8.1 Functioning

3.8.1.1 Failure to fire.-The cartridge shall not fail to fire.

3.8.1.2 Rocket Motor Functioning.-When the rocket selector cap is removed, the rocket shall fire in flight. When the rocket selector cap is left in place (set to remain inert) the rocket shall not fire.

3.8.1.3 Impact functioning.-The cartridge shall function high order on impact.

3.8.1.4 Horizontal range

3.8.1.4.1 The mean corrected horizontal range shall be within 80 meters of that predicted by the applicable range table and the standard deviation shall not be greater than 80 meters.

3.8.1.4.2 No cartridge shall have a horizontal range less than 95 percent of the mean horizontal range.

3.8.1.4.3 Muzzle velocity.-The corrected muzzle velocity of the projectile conditioned at 70 degrees Fahrenheit shall not be more than 1825 feet per second nor less than 1775 feet per second and the standard deviation of the lot shall not be greater than 8 feet per second.

3.8.1.5 Premature burst.-There shall be no premature burst.

3.8.2 Safety.-There shall be no premature burst, rocket motor malfunctioning or indication of metal parts separation in the gun bore or in flight when fired at any service charge or with propellant adjusted to produce 37,000 psi chamber pressure in a M103 or M137 cannon. No cartridge fired in the safety test shall have a horizontal range less than 95 percent of the mean horizontal range of its group.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.9 Workmanship

3.9.1 Cartridge.-All manufacturing, processing and assembly operations of the cartridge shall be correctly performed in a thorough workmanlike manner. Parts shall be free of burrs, chips, dirt, grease, rust, corrosion products and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent. Care shall be exercised at all times so that the rotating band of each motor body will not be damaged by being nicked or burred.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection providions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of loaded complete round cartridges shall contain:

a. Warhead metal parts of one lot interfix number from one manufacturer.

b. Motor body metal parts of one lot interfix number from one manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

c. Explosive charge from not more than 2 lots from one manufacturer.
d. Supplementary charges of one lot number from one manufacturer.
e. Motor body propellant grains of one lot number from one manufacturer.
f. Loaded primers of one lot interfix number from one manufacturer.
g. Cartridge cases of one lot interfix number from one manufacturer.
h. Propelling charges from one lot from one manufacturer.
i. Delay assemblies of one lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Warhead, prior to installing insulation (see dwg. 9212392 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Warhead base improperly finished ...	Visual	01001
102.	Warhead base not clean or dry	Visual	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	01003

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.2.2 Warhead, after insulation assembly (see dwg. 9212392 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Insulation assembly loose	Manual	02001
102.	Insulation and adhesive above flush	Visual	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	02003

4.2.2.3 Warhead loading assembly, prior to loading (see dwg. 9212390 covering a detail of dwg. 9212376)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Foreign matter in charge cavity (see 3.4.2)	Visual	03001
2.	Unsolidified pool of paint in bottom of cavity	Visual	03002
Major:	AQL 0.40 percent		
101.	Interior protective coating with bare spot or blister	Visual	03003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	03004

4.2.2.4 Warhead loading assembly, prior to assembly of liner (see dwg. 9212390 covering a detail of dwg. 9212376)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 percent		
101.	Depth of charge cavity	Gage	04001
102.	Foreign matter incharge cavity ...	Visual	04002
103.	Liner not coated with silicone compound	Visual	04003
Minor:	AQL 1.5 percent		
201.	Diameter of charge cavity, max.	Gage	04004
202.	Surface cavitation excessive (see 3.3.4)	Visual	04005
203.	Evidence of poor workmanship (see 3.9)	Visual	04006

4.2.2.5 Warhead loading assembly, after assembly of liner (see dwg. 9212390 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Concentricity of inside diameter of liner with pitch diameter of nose thread including perpendicularity with face of body	Gage	05001
102.	Depth to bottom of liner, min. ...	Gage	05002
Minor:	AQL 1.0 percent		
201.	Liner not expanded or can be removed by hand	Manual	05003
202.	Evidence of poor workmanship (see 3.9)	Visual	05004

4.2.2.6 Warhead loading assembly, prior to assembly of shipping plug (see dwg. 9212390 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Supplementary charge pad missing ...	Visual	06001
2.	Supplementary charge missing or inverted	Visual	06002
Major:	AQL 0.40 percent		
101.	Foreign matter in liner	Visual	06003
102.	Supplementary charge not removable by hand	Manual	06004

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Minor:	AQL 2.5 percent		
201.	Spacer missing or damaged	Visual	06005
202.	Liner missing or damaged	Visual	06006
203.	Explosive in threads (see 3.4.1)	Visual	06007
204.	Silicon compound missing from nose threads	Visual	06008
205.	Evidence of poor workmanship (see 3.9)	Visual	06009

4.2.2.7 Motor body bonding assembly prior to applying bonding material (see dwg. 9212395 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Interior of motor body not dry or containing foreign matter	Visual	07001

4.2.2.8 Motor body bonding assembly, prior to assembly of insulation (see dwg. 9212395 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Nozzle insert above flush	Visual	08001
102.	Bonding compound missing or improperly applied	Visual	08002
Minor:	None defined		

4.2.2.9 Motor body bonding assembly (see dwg. 9212395 covering a detail of dwg. 9212376)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.40 percent		
101.	Perpendicularity of insulation to motor body inside diameter	Gage	09001
102.	Excessive bonding material on outer periphery of insulation or on exposed face of insulation ...	Visual	09002
Minor:	AQL 1.0 percent		
201.	Excess bonding material on motor body (interior or exterior)	Visual	09003
202.	Painted surfaces damaged	Visual	09004

4.2.2.10 Delay housing, prior to assembly to motor body (see dwg. 9212389 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Delay assembly contents missing.....	Visual	10001
Major:	AQL 0.40 percent		
101.	O-ring missing (nozzle end)	Visual	10002
102.	Nozzle filler missing from motor body	Visual	10003
Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	10004
202.	Silicone compound not applied to O-ring and bearing surface	Visual	10005

4.2.2.11 Delay housing, after assembly to motor body and prior to assembly of selector cap (see dwg. 9212389 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Gasket missing	Visual	11001
102.	O-ring missing (selector cap end)..	Visual	11002
103.	Seal ring missing or not impressed	Visual	11003
104.	Grease on face of delay housing	Visual	11004

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	11005
202.	Silicone compound not applied to O-ring and rear face of gasket	Visual	11006

4.2.2.12 Motor body, prior to loading grain propellant (see dwg. 9212389 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.0 percent		
101.	Boron potassium nitrate pellets missing	Visual	12001
102.	Igniter closure missing, wrinkled damaged, loose or bulged	Visual	12002
103.	Selector cap missing or loose	Visual-Manual	12003
104.	Washer missing or above flush with motor body insulation	Visual	12004
105.	Propellant grain damaged	Visual	12005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	12006

4.2.2.13 Motor body, prior to assembly of warhead (see dwg. 9212389 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	O-ring missing from warhead	Visual	13001
102.	Grain cushion, warhead insulation or grain sleeve support missing or improperly assembled	Visual	13002
103.	Surface primer or sealing compound missing or improperly applied	Visual	13003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	13004

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.2.14 Projectile assembly (see dwg. 9212389 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	100 percent inspection		
101.	Excessive gap between warhead and motor body assembly	Gage	14001
Major:	AQL 0.40 percent		
102.	Projectile fails to pass through ring gage	Gage	14002
Minor:	AQL 1.5 percent		
201.	Rotating band nicked or damaged	Visual	14003
202.	Painted surfaces with total damage area in excess of 1/4 square inch	Visual	14004
203.	Marking misleading or unidentifiable including incorrect color code	Visual	14005

4.2.2.15 Primer, prior to assembly to cartridge case (see dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	One or more flash holes missing (see 3.6)	Visual	15001
Major:	None defined		
Minor:	None defined		

4.2.2.16 Propelling charge, prior to assembly to cartridge case (see dwg. 9212377 covering a detail of dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Increment missing or extra increment present	Visual	16001
102.	Any opening in bags large enough to permit escape of propellant	Visual	16002

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.0 percent		
201.	Lacing cord broken	Visual	16003
202.	Evidence of poor workmanship (see 3.9)	Visual	16004

4.2.2.17 Cartridge case, prior to assembly propelling charge (see dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush with rear of cartridge case	Gage	17001
Major:	None defined		
Minor:	AQL 1.0 percent		
201.	Primer missing or damaged	Visual	17002
202.	Foreign matter in interior of case (see 3.9)	Visual	17003

4.2.2.18 Cartridge case with propelling charge, prior to assembling to projectile (see dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Incorrect sequence of increments ..	Visual	18001
102.	Increments not properly positioned over primer	Visual	18002
Minor:	None defined		

4.2.2.19 Cartridge (see dwg. 9212376).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Shipping plug fails to pass disassembly torque test	Test	19001

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Minor:	AQL 2.5 percent		
201.	Total length, max.	Gage	19002
202.	Exterior shell with total damage area in excess of 1/4 square inch	Visual	19003
203.	Markings misleading or unidentifiable including incorrect color code	Visual	19004
204.	Cartridge case unable to shoulder on rotating band	Visual	19005
205.	Evidence of poor workmanship (see 3.9)	Visual	19006

4.2.2.20 Unsealed plastic container, when applicable (see dwg. 9209385).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Packing components missing	Visual	20001
102.	O-ring damaged	Visual	20002
Minor:	None defined		

4.2.2.21 Sealed plastic container, when applicable (see dwg. 9209385).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	O-ring not visable	Visual	21001
102.	Container damaged	Visual	21002
103.	Seal incomplete, badly wrinkled or missing	Visual	21003
Minor:	AQL 1.0 percent		
201.	Contents loose	Manual	21004
203.	Marking missing, misleading or unidentifiable, including incorrect color code	Visual	21005

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.2.22 Crate, sealed (see dwg. 9209387).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Wood broken	Visual	22001
102.	Wire or strapping missing broken or loose	Visual	22002
103.	Federal stock number or DOD identification code missing or incorrect	Visual	22003
Minor:	AQL 1.5 percent		
201.	Strapping improperly assembled	Visual	22004
202.	Car seal missing, unsealed or improperly positioned	Visual	22005
203.	Marking missing, misleading or unidentifiable	Visual	22006

4.2.2.23 Unsealed fiber container (see dwg. 9213636).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Protector tube assembly missing	Visual	23001
102.	Support ring missing	Visual	23002
Minor:	AQL 2.5 percent		
201.	Packing components improperly assembled	Visual	23003
202.	Glue or asphalt on rounds	Visual	23004
203.	Cushion missing	Visual	23005
204.	Filler between cartridge and cushion missing	Visual	23006
205.	Round cannot be removed by hand	Manual	23007

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.2.24 Sealed fiber container (see dwg. 9213636).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Container damaged or cut through asphalt layers	Visual	24001
102.	Sealing strip applied improperly	Visual	24002
103.	Metal end loose or distorted	Visual-Manual	24003
Minor:	AQL 2.5 percent		
201.	Cuts, scuffs, or gouges in outer layers	Visual	24004
202.	Tear tab length inadequate	Visual	24005
203.	Gap between body and cover in excess of 1/8 inch	Visual	24006
204.	Marking misleading, missing or unidentifiable	Visual	24007
205.	Evidence of poor workmanship (see 3.6)	Visual	24008
206.	Contents loose	Visual-Manual	24009
207.	Container cap cannot be removed by hand	Manual	24010

4.2.2.25 Sealed wooden packing box (see dwg. 9213637).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box damaged exposing contents	Visual	25001
102.	Hardware or strapping missing, broken or loose	Visual	25002
103.	Board broken	Visual	25003

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.5 percent		
201.	Strapping or hardware improperly assembled or located	Visual	25004
202.	Car seal missing, unsealed or improperly positioned	Visual	25005
203.	Marking misleading, missing, incorrect or unidentifiable	Visual	25006
204.	Handle missing or insecure	Visual-Manual	25007

4.2.3 Testing

4.2.3.1 Cavitation (see 3.3 except 3.3.4) - Critical defect - Code No. 04006.-This test shall be performed 100 percent on all warheads. Each warhead shall be subjected to x-ray examination. The test shall be performed as specified in 4.3.1 with equipment in accordance with 4.2.4. Any warhead failing to comply with the applicable requirements shall be rejected and removed from the lot.

4.2.3.2 Motor body - warhead disassembly torque (see 3.5) - Major defect - Code No. 14006.-The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-1235, using a CSP-1 plan having an AQL of 0.40 percent. Appropriate f and i values shall be as specified by the procuring activity. Any projectile which fails to comply with the applicable disassembly torque requirement shall be classed defective and removed from the lot or reworked to meet the requirements. This test shall be performed after a sealant curing time of 24 hours and in accordance with 4.3.4. Reworked projectile shall be assembled using a 275-325 foot pound torque as specified for all projectile assemblies and be retested to determine compliance with the disassembly torque requirement.

4.2.3.3 X-ray examination for missing or improperly assembled components - Major defect - Code No. 14007.-This test shall be performed 100 percent. Each assembly shall be subjected to x-ray examination in a plane that will clearly define all the components, including both columns of the delay assembly. The test shall be performed as specified in 4.3.1 with equipment in accordance with 4.2.4. Any assembly which shows evidence of missing or improperly assembled components, as listed below, shall be classed defective and removed from the lot:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

a. Delay housing assembly not seated on motor body.
b. Ring seal not fully expanded in delay and motor body grooves.
c. Incomplete boron pellet charge.
d. Propellant grain not fully seated on the motor body insulation.
e. Flash increment missing in delay column.
f. Gasket under off cap missing.
g. Rocket selector cap not fully engaged.
h. Evidence of foreign material present.
i. Any other deviation from the normal assembled conditions.

4.2.3.4 Check test for possible deterioration of primer percussion, XM108 - Code No. 15002.-If the total elapsed time between the original acceptance of the primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since the previous tests, the applicable lot shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in Specification PA-PD-2791 immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facilities assembling the complete cartridges (see 6.4).

4.2.3.5 Plastic container (when applicable).

4.2.3.5.1 Heat seal at joint (see dwg. 9209385) - Major defect - Code No. 21006.-The sampling plan for this test shall be in accordance with MIL-STD-105, Table III A, Code letter H and AQL 1.0 percent. The test shall be performed as specified in dwg. 9209385, using test equipment in accordance with 4.2.4.

4.2.3.5.2 Waterproofness of plastic container (see 3.7) - Major defect - Code No. 21007.-The sampling plan for this test shall be in accordance with MIL-STD-1235 CSP-2 Plan and AQL 0.25 percent. The test shall be performed as specified in 4.3.2 using test equipment in accordance with 4.2.4.

4.2.3.6 Proving ground.-The cartridges shall be observed for compliance with the requirements as specified in 4.2.3.6.1 and 4.2.3.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.3.6.1 Functioning.-Defects shall be classified in accordance with the following table.

Categories	Defects	Code No.
Critical	Premature burst or metal parts separation (see 3.8.1.5 and 3.8.2)	19007
	Range less than 95 percent of average range (see 3.8.1.4.2)	19008
	Rocket motor functioning (see 3.8.1.2)	19009
Major	Failure to fire (see 3.8.1.1)	19010
	Mean range (see 3.8.1.4.1)	19011
	Range standard deviation (see 3.8.1.4.1)	19012
	Muzzle velocity (see 3.8.1.4.3)	19013
	Muzzle velocity standard deviation (see 3.8.1.4.3)	19014

Minor: None defined

4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 80 sample cartridges shall be selected at random from each lot for this test, and fired with the rocket selector cap off (rocket motor to function). If two or more failures to fire occur or if a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation in horizontal range multiplied by 0.79 or the muzzle

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

velocity standard deviation multiplied by 0.79 exceeds the requirement specified in 3.8.1.4.1 or 3.8.1.4.3, or the mean range or muzzle velocity falls outside the limits specified in 3.8.1.4.1 or 3.8.1.4.3, the lot shall be rejected. The test shall be performed as specified in 4.3.3. (In the event that one or more warheads fail to show evidence of fuze functioning, additional cartridges shall be fired in order that 80 cartridges shall be observed for applicable range requirements.)

4.2.3.6.1.2 After three consecutive lots have complied with the criteria of 4.2.3.6.1.1, 50 sample cartridges shall be selected at random from each lot for this test, and fired with the rocket selector cap off (rocket motor to function). If two or more failures to fire occur or if a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation in the horizontal range multiplied by 0.75 or the muzzle velocity standard deviation multiplied by 0.75 exceeds the requirement specified in 3.8.1.4.1 or 3.8.1.4.3 or the mean range or muzzle velocity falls outside the limits specified in 3.8.1.4.1 or 3.8.1.4.3 the lot shall be rejected. The test shall be performed as specified in 4.3.3. (In the event that one or more warheads fail to show evidence of fuze functioning, additional cartridges shall be fired in order that 50 cartridges shall be observed for range requirements). If three successive lots fail to meet these requirements, the provisions of 4.2.3.6.1.1 shall be reinstated.

4.2.3.6.1.3 Impact functioning.-All rounds shall be observed for impact functioning. Firing records shall show whether the functioning was of high or low order-data will be used for informational purposes.

4.2.3.6.2 Safety.-Safety tests shall be conducted in accordance with 4.3.3.2.

4.2.3.6.2.1 Beginning with the first lot produced and continuing until three conseuctive lots have complied with the acceptance criteria specified herein, 25 sample cartridges shall be selected from each lot for this test. Twenty of the cartridges shall be fired with the rocket selector cap off (rocket motor to function) and five cartridges shall be fired with the rocket selector cap on (rocket motor not to function). If any cartridge fails to comply with any of the safety requirements specified, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.3.6.2.2 After three consecutive lots have complied with the criteria of 4.2.3.6.2.1, a sample of 13 cartridges shall be selected from each lot for this test. Ten of the cartridges shall be fired with the rocket selector cap off (rocket motor to function) and 3 cartridges shall be fired with the rocket selector cap on (rocket motor not to function). If any cartridge fails to comply with any of the safety requirements specified, the lot shall be rejected.

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-9212376 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 X-ray examination

4.3.1.1 Determination of cavities.-The warhead shall be subjected to x-ray examination (see 4.3.1.3). In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the warhead. Examination of the x-ray negative shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

made to determine compliance with the applicable requirements. In the event an original x-ray picture leaves doubt as to compliance with requirements, the warhead shall be re-x-rayed at an angle 90 degrees from that of the original exposure.

4.3.1.2 Missing or improperly assembled components.-The cartridge shall be subjected to x-ray examinations (see 4.3.1.3). In all exposures, the x-ray beam shall form a 90 degree angle with the longitudinal axis of the cartridge. Examination shall be made to determine compliance with the applicable requirements.

4.3.1.3 Radiographic equipment, operations and procedures. Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453. In addition, the sensitivity of x-ray techniques shall be 2.0 percent (4.3.1.4) and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.3.1.5.

4.3.1.4 Check of sensitivity of x-ray technique.-The sensitivity of x-ray techniques shall be determined by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the warhead or cartridge away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of the x-ray technique shall be considered satisfactory when the image of the penetrameter including the 2T hole is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on steel thickness or a combination of both charge and steel thickness, whichever applies, and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on the construction of the penetrameter, see Standard MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness to be used for the determination of cavities shall be determined as illustrated by Figure 2 of this specification.

4.3.1.5 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

portion of the material which is the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used (see 6.5). Comparison of the film with standards may be made visually, however, the use of a densitometer is recommended.

4.3.2 Waterproofness of plastic container.-The sample container shall be tested for leakage by placing it in a chamber and applying external air pressure of five (5) psi minimum (min.) to be held for fifteen seconds (min.). Observation shall be made for air pressure drop which shall be cause for rejection. The test equipment and procedure shall be in accordance with 4.2.4.

4.3.3 Proving ground tests.

4.3.3.1 Functioning.-The cartridges for this test shall be conditioned and fired at 70 degrees Fahrenheit in a howitzer equipped with a M103 or M137 cannon and a quadrant elevation of 51 degrees with a zone seven (7) propelling charge. The individual velocities and impact ranges shall be obtained and recorded. The standard deviation and the arithmetical average of the impact range shall be calculated and recorded. In addition the individual pressure, the elapsed time to rocket functioning, elapsed time to rocket burnout and velocity vs time from rocket ignition to rocket burnout shall be obtained and recorded. Radar shall also be used to observe for metal parts security and in case of short rounds or unobserved dud rounds, radar magnetic tape records of such rounds along with the three preceeding and three following rounds shall be forwarded for analysis to commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND.

4.3.3.2 Safety.-Cartridges set for rocket motor functioning shall be divided into two equal groups. Group 1 shall be conditioned at -65 degrees Fahrenheit and fired at Charge 7 in a M103 or M137 cannon. Group 2 shall be conditioned at 145 degrees Fahrenheit and fired with the propellant adjusted to give 37,000 psi chamber pressure in a M103 or M137 cannon. Group 3 cartridges with the rocket selector cap on (rocket motor not to fire), shall be conditioned at 145 degrees Fahrenheit and fired with the propellant adjusted to give 37,000 psi chamber pressure in a M103 or M137 cannon. Observation shall be made for premature burst, indication of metal part separation in the gun bore or in flight, rocket motor functioning and range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

Radar shall be used to observe for metal parts security and in case of rounds exhibiting an unusual flight, radar magnetic tape records of the round in question along with the three preceeding and the three following rounds shall be forwarded for analysis to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND.

4.3.3.3 Refiring.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional sample cartridges as required.

4.3.4 Motor body - warhead disassembly torque.-The required torque shall be applied in a loosening direction while holding the parts immediately adjacent to the joint. The joint shall be marked to provide a means of detecting any movement between the parts during the test. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Level A.-The cartridge shall be packaged in accordance with either dwg. 9209385 or dwg. 9213636, as applicable.

5.2 Packing - Level A.-The containers shall be packed in accordance with dwg. 9209387 when plastic containers are used and in accordance with dwg. 9213637 when fiber containers are used.

5.3 Marking.-When plastic containers are used the containers shall be marked in accordance with dwg. 9209385 and the wood crates shall be marked in accordance with dwg. 9209387. When fiber containers are used the containers shall be marked in accordance with dwg. 9213636 and the wood boxes shall be marked in accordance with dwg. 9213637.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army design.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format that the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5 digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1B. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness. in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Photographic densities.-"American standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, New York, has been found satisfactory as a source of information on film densities.

6.6 Clean explosive scrap.-Clean explosive scrap is defined as scrap formed in kettles, pails, risers, and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.7 Submission of ballistic test data and ammunition data cards.-In addition to the normal distribution of records for loaded items provided by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-DE and SMUPA-ND2 and SMUPA-ND4.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1315 - A - 579

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

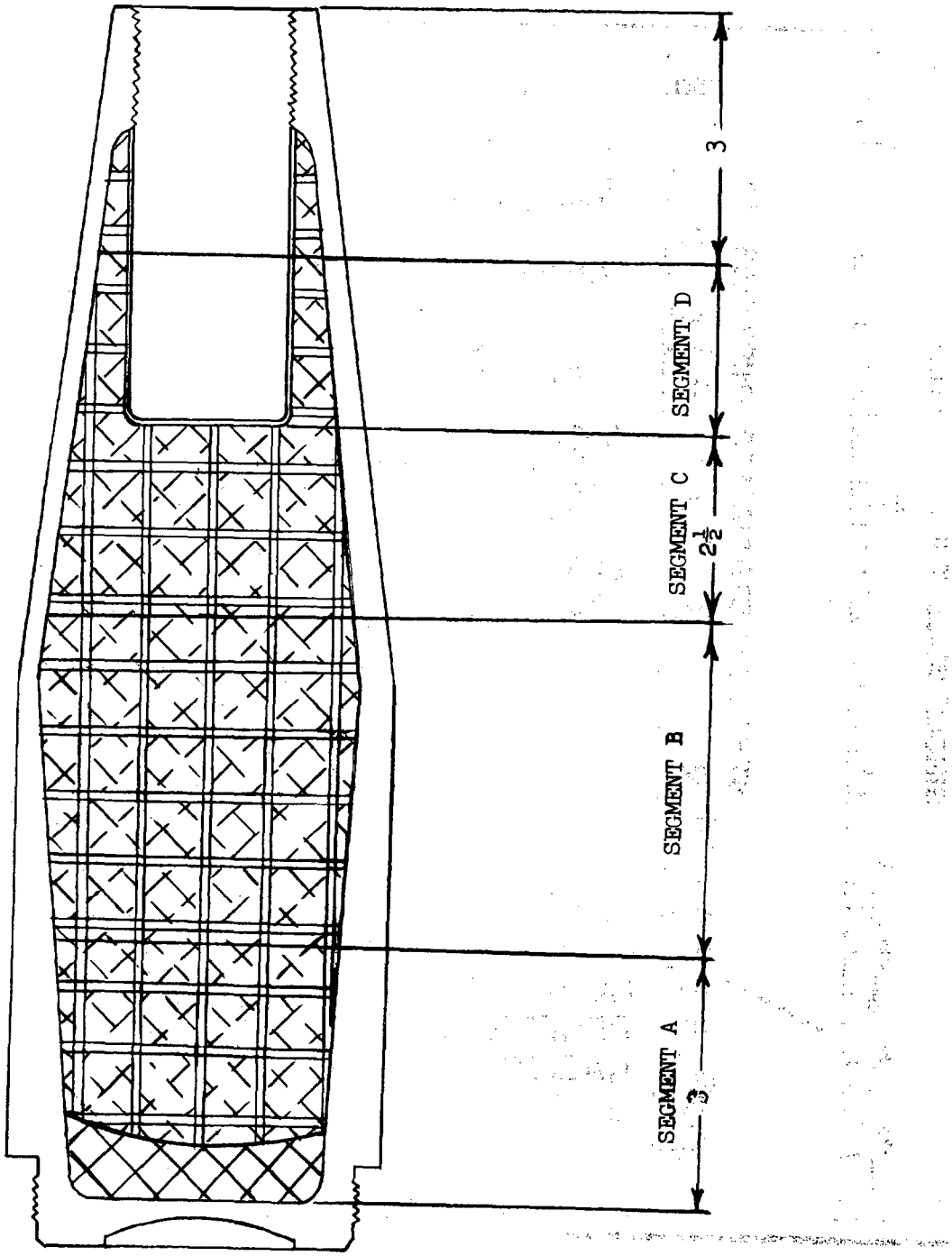
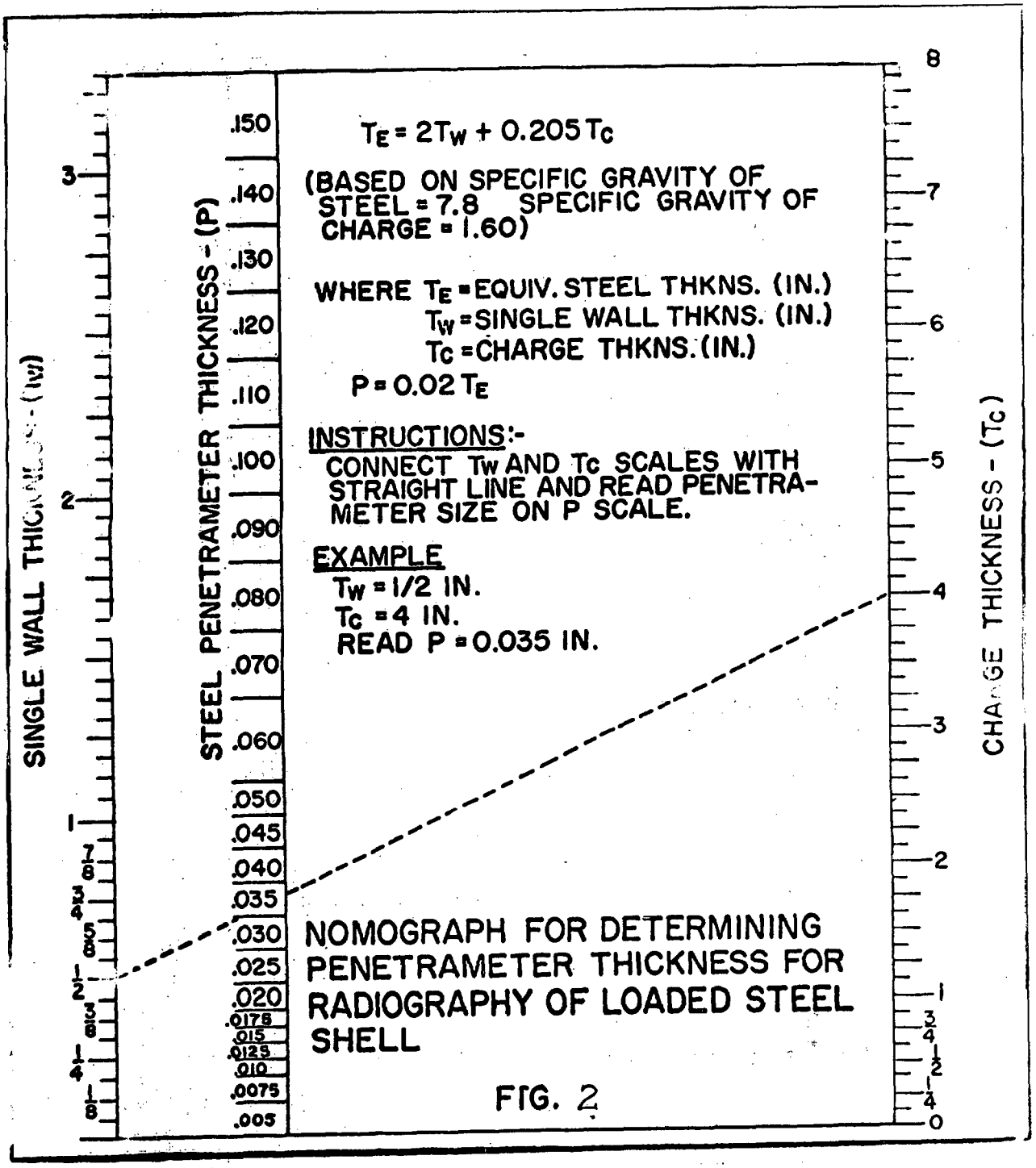


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

GPO 303-224/Z 879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.